The Nuances of Creepiness: A Systematic Literature Review of Creepy Technology

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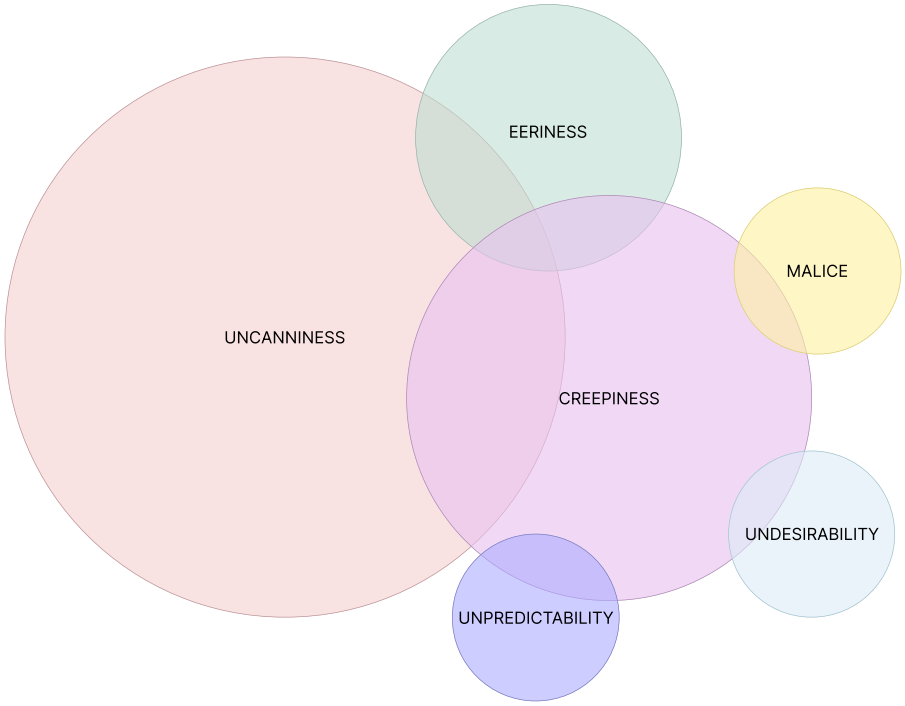


Figure 1: A Venn diagram showing the conceptual overlap among the keywords used in this literature review's search query. The diagram includes the keywords 'Uncanniness', 'Creepiness', 'Undesirability', 'Malice', 'Eeriness', and 'Unpredictability', and it highlights the particularly strong intersection between 'Uncanniness' and 'Creepiness'. The circle sizes indicate how frequently and how prominently these keywords appear in the articles in the corpus, reflecting the distribution of creepiness conceptual categories found in the reviewed literature.

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CHI '26, Barcelona, Spain

Abstract

The ambiguity surrounding 'creepy technology' in HCI poses significant challenges, as its interpretation varies considerably based

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ACM ISBN 979-8-4007-2278-3/26/04
https://doi.org/10.1145/3772318.3791030

on research focus. While it is commonly linked to the appearance of robots and concerns about data privacy and security, it is increasingly associated with functionality, unpredictability, and appropriateness of system behaviour and interaction. Clarifying this ambiguity is vital for designers, as it directly affects user experience and trust in technology. To address this, we undertook a systematic review of 115 papers, identifying key themes related to design choices and contextual elements of creepiness. Based on this review, we developed the *Creepiness Framework*, which comprises two main components: the *Structure of Creepiness* and *Felt Creepiness*. The framework clarifies the dynamics of creepy technology. Additionally, we derive design-relevant insights into creepiness features that can guide the development of technologies that are more predictable and less likely to evoke unintended feelings of creepiness.

CCS Concepts

• **Human-centered computing → Human computer interaction (HCI)**; **HCI theory, concepts and models**.

Keywords

Literature review, Meta-analysis, Literature survey, Creepy technology, Creepiness Framework, Felt Creepiness, Structure of Creepiness

ACM Reference Format:

Tora Jarsve, Barbara Sienkiewicz, Meagan B. Loerakker, Paweł W. Woźniak, and Jasmin Niess. 2026. The Nuances of Creepiness: A Systematic Literature Review of Creepy Technology. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26), April 13–17, 2026, Barcelona, Spain.* ACM, New York, NY, USA, 28 pages. https://doi.org/10.1145/3772318.3791030

1 Introduction

Creepy Technology has been extensively discussed in research, popular media and fiction. Technologies such as self-driving cars [79], smart home systems [26], and voice assistants [112], have all been described as 'creepy' in certain contexts, and the term is commonly used in media coverage of emerging technologies, for example:

> 'Amazon quietly develops creepy new technology.' [122]
>
> 'Creepy yet helpful robot is ready to assist.' [121]

As creepiness increasingly becomes a part of everyday experiences with technology, it is essential to gain a deeper understanding of these temporal and experiential nuances to better address user perceptions and improve technology design. The creepiness of technology can be a momentary experience [39], decrease over time [84], or increase [98] over time. This perception of creepiness can deter user engagement [138], raising important questions about what makes technology so unsettling and how designers can address these challenges.

In Human-Computer Interaction (HCI), research has explored creepiness in domains such as robots, avatars [46], and privacy-invading devices [91]. However, the phenomenon extends beyond any single domain. Companion technologies (like EMO, the AI Desktop pet,[1] or Amazon Echo[2]), for instance, often leverage anthropomorphism to make systems more relatable [24]. This approach can lead to unintended consequences when human-like behaviours deviate from user expectations, leading to discomfort [35]. Similarly, human augmentation technologies, such as bionic eyes or bionic limbs, can provoke unease when their proximity to the body feels intrusive or unnatural [130]. *This suggests that creepiness arises not only from technological functionality, but also from how technologies mediate interactions with users and their surrounding context.*

Existing tools such as the *Perceived Creepiness of Technology Scale (PCTS)* [138] and the SHAPE scale to measure societal perceptions toward human augmentations [131] have made strides in measuring and addressing creepiness. However, these tools often focus on specific use cases or contexts, leaving a gap in our broader understanding of how and why technologies evoke feelings of unease. For example, while some research highlights the uncanny valley [81] as a key driver of creepiness, other studies point to factors such as context [97] and social norms [90]. Addressing this complexity requires a comprehensive review that synthesises findings across multiple domains and contexts. Additionally, the absence of a unified definition or framework in the HCI literature complicates the development of cohesive design principles to address and mitigate feelings of creepiness. At the same time, given the multifaceted nature of creepiness, a singular definition may be insufficient, and diverse perspectives on what constitutes creepiness should be recognised.

Existing literature reviews (e.g. [76, 77]), highlight user discomfort. Meng-Schneider et al. [76] focuses on privacy in user interaction, while Minh Trieu and Truong Thinh [77] examine how humanoid robots' appearance and behaviour elicit unease. Both literature reviews emphasise design implications, but they primarily focus on specific domains and do not fully explore the broader complexity of creepiness across technologies and users. In particular, these studies do not address how individual factors, such as cultural background and personal experiences, contribute to perceptions of creepiness. In contrast, our literature review aims to provide an integrated understanding of creepiness within the HCI community, emphasising that it arises from both technological interactions and individual user contexts. By investigating this multifaceted nature, our review addresses a critical gap, facilitating a deeper exploration of how designers can better navigate and respond to technology-induced unease.

By addressing the gap in how creepiness in technology design is conceptualised and how such experiences emerge through design, we aim to provide a more structured understanding of how current design trends result in negative user experiences. Our goal is to offer a foundation for greater conceptual alignment in HCI, supporting efforts to avoid designs practices that risk or exploit user discomfort. At the same time, intentionally designing creepiness into technology can serve as a powerful catalyst for critical and reflective design, challenging underlying assumptions and beliefs about society and technology. In this context, the findings of our literature review can also serve as a source of inspiration for designers

[1] https://living.ai/emo/

[2] https://www.amazon.com/smart-home-devices/b?ie=UTF8&node=9818047011

and researchers engaging with creepiness deliberately and critically. To consolidate existing research, we uncover recurring patterns and identify actionable insights that benefit both interaction designers and users.

To this end, this paper aims to answer the following research questions:

> **RQ1**: How has HCI research investigated and represented the concept of 'creepy technology'?
> **RQ2**: Which design dynamics and experiential elements are linked to perceiving particular technologies as creepy?

To address these questions, we conducted a systematic literature review of research papers on 'creepy technology' published between 2019 to May 2025 through a database search on the ACM Guide to Computing Literature within the ACM Digital Library. We reviewed 6488 papers, focusing on identifying common themes and patterns related to the perception of creepiness. We analysed a final corpus of 115 papers.

Our paper makes the following contributions: 1) we provide a nuanced understanding of design and contextual factors contributing to the perception of creepiness in technology; 2) we present the *Creepiness Framework*, which provides an analytical foundation on how to develop technologies in ways that mitigate unintended user discomfort; and 3) we reflect on and discuss directions for future research on 'creepy technology', addressing both ethical implications and user experience challenges.

2 Systematic Literature Review

The goal of this literature review is to shed light on the ways 'creepy technology' has been conceptualised, designed and evaluated in HCI. We used an adapted version of the PRISMA framework [78] to report our research process. The four main phases of our systematic literature review are: (1) Identification, (2) Screening, (3) Assessment of Eligibility, and (4) Inclusion (see Figure 2). This section describes our systematic literature review process, from the identification process to our analysis.

2.1 Identification Process

To investigate how the concept of 'creepy technology' has been explored within HCI, and to identify the design characteristics and contextual factors that contribute to this perception, we conducted a systematic literature search using the ACM Digital Library[3], with the ACM Guide to Computing Literature enabled. While the ACM Digital Library provides access to the full text of ACM publications and selected third-party content, the ACM Guide to Computing Literature extends this coverage by including bibliographic records from a broader range of computing-related sources. Enabling the Guide allowed us to capture a more comprehensive set of relevant work beyond ACM's own corpus, thereby strengthening the breadth and rigour of our review.

In view of the diverse understanding of the term 'creepy technology' and related terms, we have chosen a broad approach to identify relevant literature. Our search query was designed to identify literature concerned with form-related elements or experiences associated with creepiness of technology. We searched for the terms:

> (`malic*` OR `undesirab*` OR `unpredictab*` OR `creepy` OR `eery` OR `uncanny`) OR Title:(`malic*` OR `undesirab*` OR `unpredictab*` OR `creepy` OR `eery` OR `uncanny`) OR Abstract:(`malic*` OR `undesirab*` OR `unpredictab*` OR `creepy` OR `eery` OR `uncanny`).

Terms such as malice, undesirability and unpredictability were included due to their role in the conceptual model of creepiness by Woźniak et al. [138], while creepy, eerie and uncanny were included as commonly used descriptors of unsettling or ambiguous experiences with technology.

The initial query yielded 16,481 results published between 1952 and May 2025. We included full papers, short papers, pictorials and work-in-progress contributions (also called late-breaking work or extended abstracts). Full papers typically provide in-depth empirical, artefact or theoretical contributions, while shorter formats often present early-stage insights or speculative designs. Pictorials were particularly relevant for surfacing experiential or aesthetic dimensions of creepiness in design.

Rather than limiting our scope to specific venues or formats, we applied content-based inclusion criteria across all publication types. We focused on content-based criteria, as research often treats creepiness implicitly through related or synonymous terms; hence, extending our search to a wide range of paper types provided us with the opportunity to attain comprehensive knowledge of how theory and practice influence the perception of creepy technology. In line with this, applying no limitations on publication venues yielded papers with different kinds of research foci, including design, empirical, and theoretical works—allowing us to review creepiness from different research perspectives. Table 6 and Figure 7 in the Appendix provide an overview of the publication venues of the papers included in the corpus. They show that there is a relatively wide distribution of publication venues, as most venues are present only a single time in the corpus (32 out of 115, 27.83%). Most papers were published in CHI (23 out of 115, 20%), followed by CiHB (10 out of 115, 8.70%), HRI (8 out of 115, 6.96%), IJHCS (7 out of 115, 6.09%), and IEEE Trans. Vis. Comput. Graph. (5 out of 115, 4.35%).

To reflect current trends in HCI, we excluded papers published prior to 2019. Our review focuses on research published from 2019 onward to build an understanding of how recent work engages with creepy technology. Earlier contributions form an important foundation for this area. However, we chose to concentrate on this more recent period to develop a clearer view of the ideas shaping current discussions. This filtering process resulted in a screening set of 6,488 papers. The selection process is visualised in Figure 2.

2.2 Screening Process

The first and second authors independently screened 6,488 papers based on titles and abstracts. Papers were excluded if they did not address user interactions with or the design of 'creepy technology', or if they did not report empirical accounts of unsettling user experiences. Studies lacking relevance to user perceptions of creepiness, for example, those focused solely on collaboration or task

[3] https://dl.acm.org/

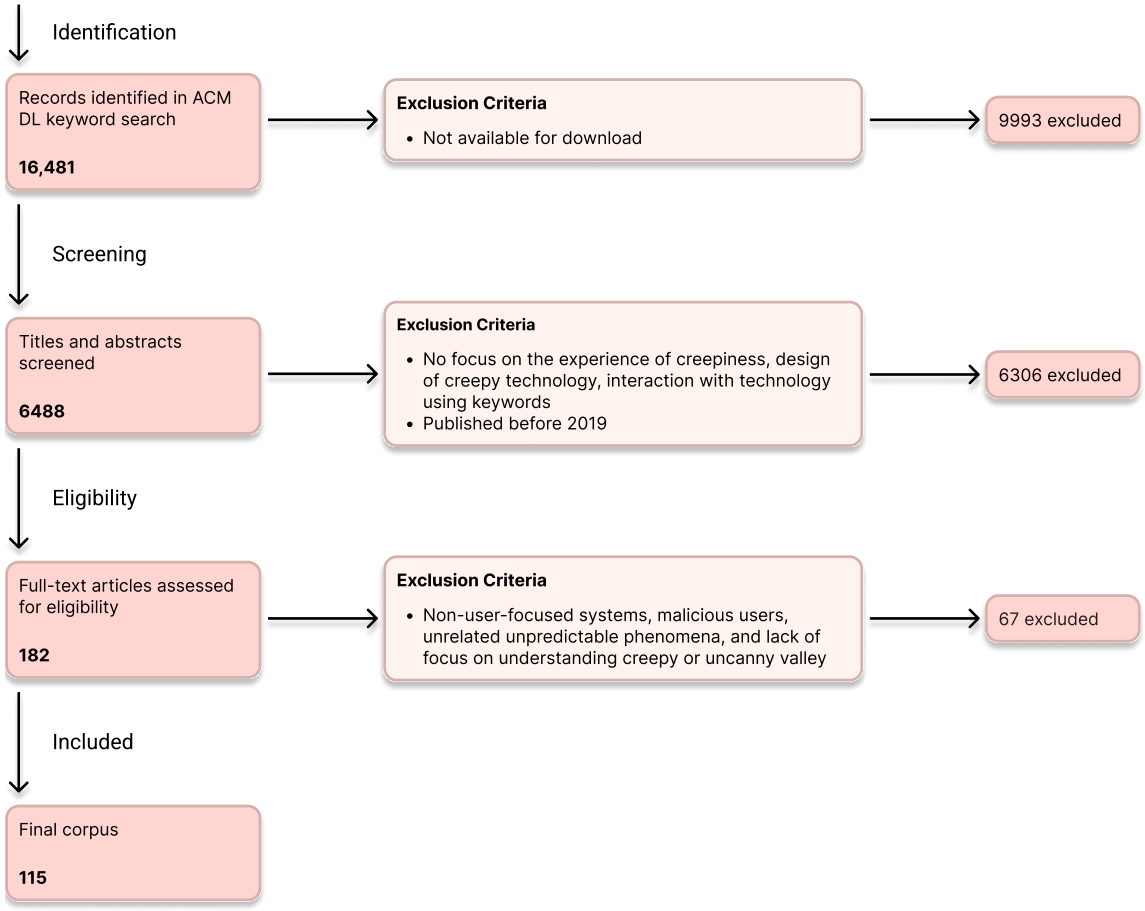


Figure 2: Adapted PRISMA flow diagram reporting the selection process, from initial identification to the final inclusion of papers in our literature review.

performance between humans and non-human agents, were omitted. Moreover, papers that focused solely on the technical aspects of using technology in a malicious way or on how to prevent it, were also omitted, for example Hörnemann et al. [49].

We also excluded work on *dark patterns*, as these reflect designer intent rather than intrinsic technological features. Our aim was to identify characteristics of technology itself that elicit creepiness, rather than broader manipulative design strategies [2]. This process resulted in the exclusion of 6,306 papers. A total of 182 papers were retained for the eligibility assessment.

2.3 Assessing Eligibility

The first and second authors independently reviewed 105 and 77 papers, marking those for which the relevance to 'creepy technology' was unclear. Iterative discussion sessions were held between the first, second and last authors throughout this process to review the marked papers.

Papers were excluded if they did not include interaction or embodiment-related elements, report empirical studies, or reflect on the human experience with potentially 'creepy technologies'. Moreover, papers that referenced the keywords only as antonyms or contrasts to the constructs of interest, for example, mentioning creepiness solely as the opposite of creating trust, were also excluded. This step resulted in the exclusion of 67 additional papers, leaving a final set of 115 papers.

2.4 The Final Corpus

The remaining 115 papers were divided into five sets and distributed among five authors. Each author read their assigned papers in full and assessed them against the exclusion criteria outlined in Figure 2. At this stage, the evaluation focused solely on applying the exclusion criteria; no additional analysis was conducted. Papers that raised uncertainty regarding exclusion were flagged for group discussion. Final decisions on inclusion or exclusion were made collectively by the first, second and last author. Notably, no additional papers were excluded during this step, resulting in a final corpus of 115 included papers.

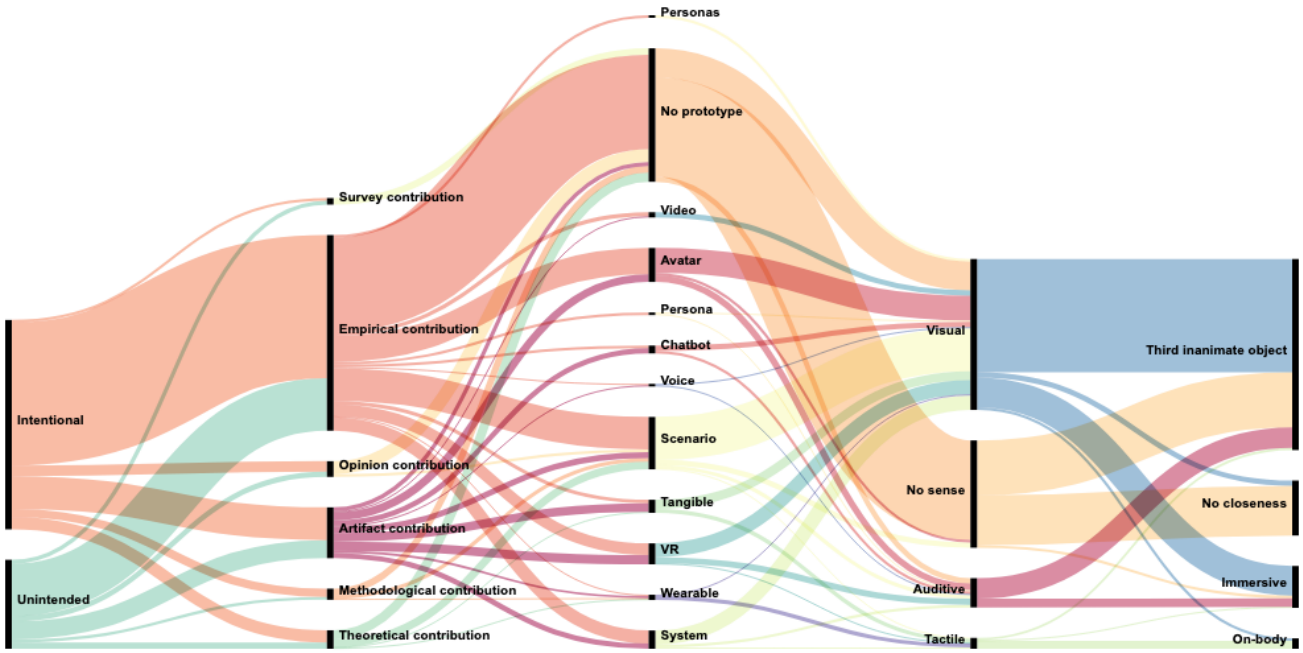


Figure 3: A Sankey diagram illustrating the relationships between various coding categories used in the analysis to demonstrate to what extent categories overlap, highlighting potential gaps in research on creepy technology. From left to right, the following coding categories are visualised: creepiness `intention`, `type of contribution`, `prototype type`, `sensory modality`, and `proximity to the body`.

2.5 Analysis

In our analysis, we applied a consensus-based approach [12]. In line with this, no inter-rater reliability was calculated. This approach is similar to what Stefanidi et al. [123] and Kaltenhauser et al. [56] also adopted in their literature reviews. To address our research questions and explore the concept of 'creepy technology', we utilised a multi-phase analytical approach. The initial phase involved developing a comprehensive coding scheme. The first and last authors independently reviewed a shared research paper and generated preliminary codes separately. Subsequently, they engaged in collaborative discussions to reconcile and refine their codes. This iterative process was repeated four times to ensure the coding scheme's refinement. In the next phase, the first and second authors analysed eight randomly selected papers from the corpus, further developing the coding tree. By separately coding and then discussing discrepancies, they resolved disagreements and refined the coding tree, including detailed descriptions for clarity. Once all papers had been coded, the first and second authors consolidated the codes to streamline the subsequent analysis. Figure 3 provides an overview of how the codes are connected.

The papers were coded in five metacategories: **basic bibliographic information** (title, author, DOI, publication venue, and year), which can be directly obtained from the reference, and four additional metacategories that are described in Table 1.

We examined how and when the **concept of creepiness** or related keywords appeared in each paper. This included determining whether the authors had the `intention` to evoke feelings of creepiness or design a creepy technology. If they explicitly focused on creepiness (or related concepts such as the uncanny valley), we categorised it as INTENTIONAL CREEPINESS. If it emerged as a secondary finding during the research, we categorised it as UNINTENTIONAL CREEPINESS. We also analysed the `definitions` used in the papers and identified key factors in the definitions that contribute to creepiness.

Empirical research details (**study details**) were also coded, including the `study setting`, `study duration` (cross-sectional or longitudinal), `sample size`, `types of participants` (e.g. adults, students, children), `data collection methods`, and the overall `research approach` (qualitative, quantitative, or mixed methods). To provide a comprehensive overview of how prototypes were used in the literature, we also analysed whether prototypes were developed during the research. Following the analysis of each prototype, we applied additional codes, including the `data collection methods` for ideation purposes, the method of prototyping (`prototype development`), the creator of the prototype (`prototype developer`), the types of technologies used for the prototypes (`existing technologies as prototypes`), the `method of evaluation`, the prototype's `tangibility`, `sensory modality`, and `proximity to the body`. As creating a prototype is a formative contribution, we annotated the papers in which the authors described a prototype being developed. Given that the reporting of the prototyping process was somewhat fragmented, we chose to conduct a descriptive rather than a quantitative analysis.

Table 1: Overview of the coding categories applied in the analysis. Categories were clustered into broader metacategories: the paper's research focus, the concept of creepiness, empirical research details (study design), and prototypes.

Metacategory	Category	Values
Research focus	Type of paper	Full, pictorial, extended abstract
	Main research aim	*Bottom-up qualitative coding*
	Research questions	Binary: Yes/no; if yes, *bottom-up qualitative coding*
	Type of contribution	Empirical, artefact, methodological, theoretical, dataset, survey, opinion (following the categorisation of Wobbrock and Kientz [137])
Concept of creepiness	Intention	Intentional, unintentional
	Definitions	Explicit, paraphrasing, connecting, contextual, no definition
Study design	Study setting	Lab, field, online, unspecified, not applicable
	Study duration	Cross-sectional, longitudinal, not applicable
	Sample size	Number of participants engaged in the research
	Types of participants	Students, adults, children, Artificial Intelligence, unspecified, not applicable
	Data collection methods	Interview, demo, workshop, EEG, questionnaire, focus group, web content analysis, somatic reflections, observations, literature review, experiment, keywords, exclusion, eye tracking, mouse tracking, Ethnomethodological Conversation Analysis (EMCA), video observations, not applicable
	Research approach	Qualitative, quantitative, mixed methods, not applicable
Prototypes	Prototype development	Low-fidelity, multimedia and animation, virtual environment and digital spaces, wearables, physical artefact, not applicable
	Existing technologies as prototypes	Videos, images, text, physical robots, digital robots, not applicable
	Prototype developer	Researchers, participants, not applicable
	Method of evaluation	Binary: Yes/no; if yes, *bottom-up qualitative coding*
	Tangibility	Tangible, non-tangible, not applicable
	Sensory modality	Taste, tactile, visual, auditive, olfactory
	Proximity to the body	In-body, on-body, third inanimate object, immersive, not applicable

Table 2: The applicable coding categories for each paper in our corpus. The meaning of each sign is as follows: •-Intentional, ○-Unintentional, F-Full Paper, S-Short Paper and Extended Abstract, P-Pictorial, □-No, ■-Yes, 𝔸-Artefact, 𝔼-Empirical, 𝕄-Methodological, 𝕆-Opinion, 𝕊-Survey, 𝕋-Theoretical, Quant-Quantitative, Qual-Qualitative, MM-Mixed Methods, U-Digital Robot, §-Video, ¶-Image, ±-Text, ÷-Physical Robot, ⊗-Virtual Environments and Digital Spaces, ◂-Low-fidelity, △-Physical Artefact, 𝔪-Multimedia and Animation, ω-Wearables, V-Visual, A-Auditive, T-Tactile, ♠-Immersive, ♡-Third Inanimate Object, ♣-On-body.

Authors	Intent focus	Paper type	RQ	Contribution	Approach	Existing tech.	Prototype dev.	Tangibility	Senses	Proximity
Al-Zubeidi et al. [1]	•	F	□	𝔸			⊗	□	V A	♠
Alimardani et al. [3]	•	F	■	𝔼	Quant	U				♡
Appel et al. [4]	•	F	□	𝔼	Quant		◂	□	V	♡
Ashby et al. [5]	○	F	□	𝔼 𝔸	MM		◂ △	■	V	♡
Ashok et al. [6]	○	S	□	𝔼	MM	§	◂	□	V A	♡
Aylett et al. [7]	•	F	□	𝕆		U				
Babel et al. [8]	•	S	□	𝔼 𝕄	Quant	¶				♡
Baek and Kim [9]	•	F	□	𝔼 𝕋	MM					
Bakan et al. [10]	•	F	□	𝔼	Quant		𝔪	□	V A	♡
Borem et al. [13]	•	F	■	𝔼 𝔸	Qual		⊗	□		♡
Brideau-Duquette et al. [14]	•	F	□	𝔼	Quant		⊗	□	V	♠
Brunswicker et al. [15]	○	F	■	𝔼	Quant		◂	□	V	♡
Cerezo et al. [16]	○	F	□	𝔸	MM		⊗	□	V	♡
Chen et al. [17]	•	S	□	𝔸						
Chen and DiFranzo [19]	•	F	■	𝔼	Quant		⊗	□	V A	♡
Ciechanowski et al. [20]	•	F	□	𝔼 𝔸	MM		⊗	□	V A	♡
Collyer-Hoar et al. [21]	○	F	□	𝔼	MM		◂	□	V	♡
Combe et al. [22]	•	F	■	𝔼 𝔸	Quant	U	⊗	□	V A	♠
Conner et al. [23]	•	S	□	𝕆						
Kats and Sharif [58]	○	F	□	𝔼	MM					
Higgins et al. [44]	•	F	□	𝔼 𝔸	Quant		⊗	□	V A	♡
Demers and Vorn [25]	•	S	□	𝔼 𝔸			⊗	■	T V	♣
Desnoyers-Stewart et al. [27]	○	F	□	𝔼 𝔸	Qual		⊗	□	T V A	♠
Diel et al. [29]	•	F	□							
Eid et al. [30]	•	S	□	𝔼 𝔸	Quant	§				♡
Fiolka et al. [31]	•	S	□	𝔼	Quant	¶				♡
Frampton-Clerk and Oyekoya [32]	○	F	□	𝔼	Quant		⊗	□	V A	♡
Grundke et al. [35]	•	F	□	𝔼	Quant	§				♡
Guo et al. [36]	○	F	■	𝔼	MM		⊗	□	V A	♠
Hanson et al. [37]	○	F	□	𝔼	MM		𝔪	□	V	♡
Hao et al. [38]	•	S	□	𝔸	Quant		𝔪	□	V	♡
Hatada et al. [41]	○	F	□	𝔼 𝔸	MM		⊗	□	V	♠
Hepperle et al. [42]	•	F	□	𝔼	Quant	¶				♡
Herder and Zhang [43]	•	F	■	𝔼 𝕋	MM	±				
Higgins et al. [45]	•	F	□	𝔼	Quant	¶				
Hoorn and Huang [47]	•	F	■	𝔼 𝕄	Quant	÷U				♡
Hover et al. [50]	•	F	□	𝔼 𝕋	Quant	±				
Husca [52]	•	S	□	𝕊						
Hartbrich et al. [40]	•	F	□	𝔼 𝔸	Quant		⊗	□	V	♠
Jarsve et al. [54]	•	S	□	𝔼	MM	÷U				♡
Kühne et al. [62]	•	F	□	𝔼	MM	¶			V	♡
Karpashevich et al. [57]	○	F	□	𝔸 𝕄 𝕋	Qual		ω	■	T	♣

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Table 2 – continued from previous page

Authors	Intent focus	Paper type	RQ	Contribution	Approach	Existing tech.	Prototype dev.	Tangibility	Senses	Proximity
Kenny et al. [59]	●	F	□	𝔼 𝔸	Quant		⊗	□	V	♡
Kim et al. [60]	●	F	□	𝔼	Quant	¶				♡
Kleinbeck et al. [61]	●	F	■	𝔼	MM		⊗	□	V A	♠
Kullmann et al. [63]	●	F	■	𝔼	MM	÷		□	V	♠
Kaate et al. [55]	●	F	■	𝔼	Quant		◂	□	V A	♡
Lavado-Nalvaiz et al. [64]	○	F	□	𝔼 𝕋	Quant	÷				
Lavi et al. [65]	●	P	□	𝕆	Quant	¶				♡
Lee et al. [66]	○	F	■	𝔼 𝔸 𝕋	MM		△	■	V	♡
Li et al. [67]	●	F	■	𝔼 𝕋	Quant	U				♡
Löffler et al. [68]	●	F	□	𝔼	Quant					
Severgnini et al. [111]	●	F	□	𝔸	Quant		ω	■	T	♣
MacDorman [70]	●	F	□	𝔼 𝔸	MM		⊗	□	V A	♡
Macias-Huerta et al. [71]	●	F	□	𝔼	Quant	U	⊗	□	V	♡
Mal et al. [72]	●	S	■	𝔼 𝔸	Quant		⊗	□	V	♡ ♠
Mathur et al. [73]	●	F	□	𝔼	Quant	¶		□	V	♡
Mazuz and Yamazaki [74]	○	F	□	𝔼𝕋	Qual	÷				♡
McDonald [75]	○	F	□	𝔼𝔸	MM		⊗	□	V	♡
Sharma and Vemuri [116]	●	F	■	𝔼	Quant	§¶U				
Meng-Schneider et al. [76]	○	S	■	𝕄𝕊						
Minh Trieu and Truong Thinh [77]	○	F	□	𝕊						
Saisubramanian et al. [102]	●	F	□	𝔼	Quant	÷	m	□	V	♡
Muralidhar et al. [82]	●	S	□	𝔼	Quant					♡
Nissen et al. [83]	●	F	□	𝔼	Quant	¶				♡
Benjamin et al. [11]	○	F	□	𝕆						
Paetzel et al. [87]	●	F	■	𝔼	Quant	÷				♡
Paetzel-Prüsmann et al. [88]	●	F	■	𝔼	Quant	÷				♡
Paetzel and Castellano [86]	●	F	□	𝔼	Quant		m	□	V A	♡
Parviainen and Søndergaard [89]	●	F	□	𝔼𝔸	Qual		△	■	T V A	♡
Phinnemore et al. [90]	●	F	□	𝔼𝕄𝕋	MM		◂	□	V	♡
Raff et al. [92]	●	F	■	𝔼𝕄	MM					
Rapp et al. [93]	●	F	■	𝔼	Qual	U		□	V	♡
Rapp et al. [94]	○	F	■	𝔼	Qual		m	□	V	♡
Gahrn-Andersen [33]	●	F	□	𝕋						
Ratajczyk [95]	●	F	□	𝔼	Quant			□	V	♡
Register et al. [96]	○	F	■	𝔼	Qual	±				
Reig et al. [97]	●	F	□	𝔼𝕄	Qual		◂	□	V	♡
Reinhardt et al. [99]	○	F	■	𝔼𝔸	MM		⊗	□	V A	♠
Rudaz et al. [100]	○	F	□	𝔼	MM	÷				♡
Sabinson and Green [101]	○	F	□	𝔼𝔸	MM		△	■	V	♡
Salminen et al. [103]	●	F	■	𝔼	Quant		◂	□	V	♡
Sandhaus et al. [104]	●	S	■	𝔼 𝔸	Quant		⊗	□	V	♡
Saneyoshi et al. [105]	●	F	□	𝔼	Quant	¶				♡
Savery et al. [106]	○	F	□	𝔼 𝔸	MM		m	□	V A	♡
Schweinberger et al. [107]	●	F	□	𝔼	Quant	§				♡
Scott et al. [108]	●	F	■	𝔼	Qual	§±				♡
Seberger et al. [110]	●	F	■	𝔼 𝕄	Quant					
Seberger et al. [109]	●	F	■	𝔼 𝕋	Qual		◂	□	V	♡
Shaheen et al. [113]	○	F	□	𝔼 𝕋	Qual	U				♡
Shalawadi et al. [114]	●	F	□	𝔼 𝔸	MM		△	□	V	♡

Continued on next page

Table 2 – continued from previous page

Authors	Intent focus	Paper type	RQ	Contribution	Approach	Existing tech.	Prototype dev.	Tangibility	Senses	Proximity
Chen et al. [18]	●	F	■	𝔼	Quant		𝔪	□	V	♡
Sharevski et al. [115]	●	S	□	𝔼 𝔸	Quant		⊗	□	V	♡
Shih et al. [117]	●	F	■	𝔼	MM		⊗	□	V	♠
Shin et al. [118]	●	F	■	𝔼 𝔸	Quant		⊗		V	♠
Shklovski and Grönvall [119]	●	F	□	𝔼 𝔸	Qual		ω	■	T V	♣
Sixsmith [120]	●	F	□	𝕆						
Strait et al. [124]	●	F	□	𝔼	Quant	¶				♡
Tanaka et al. [125]	○	F	□	𝔸	Quant		△	■	T V	♡
Torkamaan et al. [126]	●	S	■	𝔼	Quant		◂	□	V	♡
Tseng et al. [128]	○	F	□	𝔼 𝔸 𝕄 𝕋	MM		◂	□	V	♡
Turner et al. [129]	●	S	□	𝔼	Quant	¶§		□	V A	♡
Wang et al. [133]	○	F	□	𝔼	MM		⊗	□	V A	♠
Wang et al. [132]	○	S	■	𝔼 𝔸	MM		⊗	□	V A	♡
Wang et al. [134]	●	F	■	𝔼	MM	U		□	V	♡
Wiese and Weis [135]	●	F	□	𝔼	Quant		𝔪	□	V	♡
Wisessing and McDonnell [136]	○	S	□	𝔼	Quant		⊗	□	V	♡
Woźniak et al. [138]	●	F	□	𝔼 𝕄 𝕋	MM	¶	◂	□		♡
Yam et al. [139]	●	F	□	𝔼	Quant	÷				♡
Yamaguchi [140]	●	F	□	𝔼	Quant	¶		□	V	♡
Yang et al. [141]	●	F	■	𝔼 𝔸	MM		⊗	□	V A	♠
Yip et al. [142]	●	F	■	𝔼 𝕋	Qual	§	◂	□	V	♡
You et al. [143]	○	F	■	𝔼	Quant		⊗	□	V	♠
Yurman and Reddy [144]	○	F	□	𝕆	Qual		◂	□	V	♡
Zhang et al. [145]	○	F	■	𝔼	Qual					
Summarised	●=81	F=95	□=76	𝔸=34 𝔼=97	Quant=55	U=11	⊗=29	□=62	V=71	♠=16
	○=34	S=19	■=38	𝕄=10 𝕆=6	Qual=16	§=8	◂=16	■=9	A=23	♡=74
		P=1		𝕊=3 𝕋=15	MM=32	¶=15	△=6		T=7	♣=4
						±=4	𝔪=11			
						÷=10	ω=3			

3 Results

In this section, we report the findings of our analysis in relation to our first research question:

- **RQ1**: How has HCI research investigated and represented the concept of 'creepy technology'?

Table 2 presents a comprehensive overview of the coding categories applied to the papers in our corpus. These categories capture how each paper engaged with 'creepy technology'. Table 2 showcases several prominent patterns in the corpus, including the fact that the majority of records were full papers, studied creepiness INTENTIONALLY, had concrete research questions, had a TANGIBLE prototype, primarily utilised VISION as `sensory modality` in the prototype(s), and made an EMPIRICAL contribution. QUANTITATIVE research approaches were most common in the corpus, though MIXED METHODS studies were also utilised frequently. Regarding `prototype development` practices, the majority of prototypes fell in the category of VIRTUAL ENVIRONMENTS AND DIGITAL SPACES; though, a good number of prototypes were LOW-FIDELITY or utilised MULTIMEDIA AND ANIMATIONS. We observed a lack of diversity regarding the prototypes' `proximity to the body`, as the far majority of prototypes were THIRD INANIMATE OBJECTS.

3.1 Defining Creepiness and Related Concepts

To examine how creepiness and related concepts are defined in HCI, we focused on papers that explicitly engaged with these terms in our search query. Through iterative analysis, we identified five distinct approaches to creepiness `definitions`: (i) PARAPHRASING, where authors reworded existing definitions without direct attribution rather than proposing their own; (ii) CONNECTING DEFINITIONS, which linked terms directly to specific examples within the research narrative, typically by situating creepiness in concrete study contexts; (iii) NO DEFINITION, where terms were used without clarification or explanation; (iv) CONTEXTUAL DEFINITIONS, which inferred meaning through empirical characteristics, models, or observed incidents rather than through direct statements; and (v) EXPLICIT AND CLEAR DEFINITIONS, characterised by authoritative statements such as "X is defined as…", often with supporting citations. Paraphrasing was the most common approach. Overall, 87 papers (75.6%) provided some form of definition, while 28 (24.3%) did not (see Table 3).

Table 3: Distribution of papers across the different types of `definition` categories. Note that some papers fell into more than one category, with the exception of those that fell into the `no definition` category.

Definition category	Number of papers
Paraphrasing	70
Connecting definitions	4
No definition	28
Contextual definitions	26
Explicit and Clear definitions	5

The analysis of definitions began with collecting them all on a Miro[4] board, they were then subsequently segmented into smaller units to enable more detailed analysis. From these units, keywords were systematically extracted and assigned colour codes corresponding to search terms such as creepy, malicious, unpredictable, undesirability, eerie, and uncanny valley. The keywords were then organised according to their referents, clustered based on thematic similarity, and finally arranged to illustrate areas of conceptual overlap between clusters (search terms). Each step was initially conducted independently by the first and second authors, after which the results were discussed collaboratively to reduce bias. We held collaborative discussions to reconcile any differing views on the clustering, ensuring that our categorisations were as accurate as possible. One of the outcomes of this analysis is the Venn diagram presented in Figure 1. The size and overlap of the clusters correspond to the overlap of the definitions, while singular units were removed to improve readability. Table 4 quantifies the frequency of each definition keyword in the literature, with 'Uncanny' being the most prevalent at 66.9%, while 'Undesirability' is the least common at 1.7%. This complements the Venn diagram visualisation (Figure 1) by showing the relative emphasis of each conceptual category in existing research.

3.1.1 Paraphrasing. PARAPHRASING was the most commonly used approach among the papers we analysed, appearing in 70 papers (60.86%). Paraphrasing entails rewording existing definitions from the literature without direct attribution. A typical example is the frequent use of the concept of the 'Uncanny Valley', as described by Mori et al. [81], although paraphrasing was also applied to other concepts related to creepiness. The Uncanny Valley effect posits that as a robot's or artificial agent's human likeness increases, people's affinity towards it increases as well—until a point at which near-human appearance, movement, or behaviour elicits feelings of eeriness, revulsion, or discomfort, creating a dip (the 'uncanny valley') in affinity as human likeness increases towards realism. The paraphrasing approach allows authors to convey established concepts in their own words while retaining the fundamental meaning.

3.1.2 No Definition. Papers in the NO DEFINITION category used key terms without offering clarification or explanation. To illustrate, in these cases, concepts such as 'unpredictable' or 'undesirability' were employed as if self-evident, with no accompanying description of their meaning. A total of 28 papers (24.35%) fell into this category.

3.1.3 Contextual Definition. CONTEXTUAL DEFINITIONS were identified in 26 papers (22.61%) and are utilised to explain terms through inference rather than direct statements, often without supporting citations. This category bridges the gap between explicit definitions and paraphrasing by providing meaning through contextual nuances. CONTEXTUAL DEFINITIONS are characterised by their reliance on the specific context of the work to elucidate terms. They are typically embedded within the study's empirical findings, drawing on characteristics, models, or observed incidents to infer the meaning of concepts like 'creepiness' indirectly.

For example, Herder and Zhang [43, p. 4] contextually define creepiness by noting that 'the overwhelming majority of creepy ad incidents (17 out of 19) were attributed to unacceptable/unexpected

[4] https://miro.com/index/

Table 4: Frequency of conceptual definition categories identified in the corpus. Eighty-seven papers defined creepiness using keywords such as 'uncanniness', 'creepiness', 'undesirability', 'malice', 'eeriness', 'unpredictability', and their conjugated forms. As papers often employed multiple keywords to characterise creepiness, categories overlap to varying degrees in the Venn diagram (Figure 1), and the "# Papers" column therefore sums to more than 87.

Conceptual definition category	# Papers	Percentage	Example
Creepy/Creepiness	19	16.52%	'[...] we define "creepiness" as a feeling of discomfort that is experienced when a user is not sure how to interact with AI.' [9, p. 3]
Malice/Malicious	3	2.6%	'[...] 'implied malice' (*'intention or desire to do evil or cause injury'* [28]) [...]' [138, p. 5]
Unpredictability	4	4.7%	'[...] *unpredictability* (i.e., an ability to predict the actions of a technology or having a lack of control over a technology) [...]' [90, p. 3]
Undesirability	2	1.7%	'*Undesirability* in our model refers to users perceiving the interactive technology as a non sequitur; a feeling of unease caused by the interactive artefact being out of context' [138, p. 5]
Eerie	8	6.9%	'The uncanny valley is a phenomenon in which as robots become more human-like, people see them as strange and weird [80, 81].' [142, p. 3]
Uncanny (incl. Uncanny Valley of Mind)	77	66.9%	'In design, the uncanny valley is often considered as the hypothesised relationship between an entity's resemblance to a human being (realism) and the users' emotional responses to the entity.' [103, p. 10]
Total	**87**	**100%**	

data collection practices.' Herder and Zhang [43] do not provide a direct definition of 'creepiness' but infer it from recurring features observed in specific cases. Similarly, Woźniak et al. [138, p. 5] discuss their 'model of creepiness', which consists of three dimensions: implied malice, undesirability, and unpredictability. While Woźniak et al. [138] clearly articulate what these dimensions entail, the model remains conceptual, relying on the interplay of these dimensions to suggest a definition of creepiness, rather than adopting established criteria or citing prior sources. Importantly, this choice is intentional, as the authors' aim was to establish a novel conceptual understanding of creepiness grounded in their empirical findings, rather than to draw on existing definitions from other fields. Likewise, Yip et al. [142] offer a contextual definition of creepiness by highlighting five design properties that signal creepiness in technology: 1) deception vs. transparency, 2) ominous physical appearance, 3) lack of control, 4) unpredictability, and 5) mimicry. These empirically grounded properties outline aspects to consider, forming an interpretive framework anchored in a research context rather than a direct or traditional definition. As with Woźniak et al. [138], this approach was deliberately adopted to foreground a new conceptualisation of creepiness, rather than importing definitions established in adjacent research domains.

3.1.4 Connecting Definition. This category is somewhat similar to contextual definitions. The key difference is that contextual definitions are constructed from the empirical characteristics in the research, often indirectly, while CONNECTING DEFINITIONS involve tying the concepts directly to specific contextual examples within the discourse, making them more explicitly part of the written narrative rather than constituting formal study findings. Unlike findings that could become part of a framework such as the one by Woźniak et al. [138], CONNECTING DEFINITIONS tie concepts directly to the research context. For instance, Parviainen and Søndergaard [89, p. 9] link their definition of creepiness directly to their study on a whispering voice assistant: 'One of the experiential qualities of interacting with a voice assistant through whispering is a feeling of creepiness. The assistant can be perceived as creepy because the voice modality of whispering breaks with our social norms and expectations of voice interaction.' This allows researchers to ground abstract concepts in concrete examples from their empirical work. Four papers (3.48%) fell into this category.

3.1.5 Explicit and Clear Definition. Across the papers we reviewed, only a small subset provided EXPLICIT DEFINITIONS of creepiness. In total, 5 out of 115 papers (4.35%) used this approach. EXPLICIT DEFINITIONS are characterised by direct statements that articulate what is meant by a term. These can range from highly formal constructions, such as *'X is defined as…'*, to more general but still deliberate formulations that clearly convey a working definition. This method offers clarity for the reader and often anchors the concept within a specific context or disciplinary perspective.

For example, Baek and Kim [9, p. 3] state: *'We define "creepiness" as a feeling of discomfort that is experienced when a user is not sure how to interact with AI.'* Similarly, Seberger et al. [110] describe creepiness as 'emotional discomfort,' thereby situating the term within affective experience. These types of definitions, whether formally stated or clearly articulated, provide a consistent conceptual foundation and help delineate how creepiness is understood within each study.

3.2 Intentional versus Unintentional Creepiness

Our analysis distinguished between two types of contributions based on the `intention`: those that intentionally investigated creepy technology (INTENTIONAL CREEPINESS) and those in which creepiness emerged as an unintended outcome (UNINTENTIONAL CREEPINESS). This distinction is critical for understanding how creepiness functions in interaction design contexts: while some studies set out to probe creepiness as a design or experiential quality, others revealed creepiness incidentally, often as a by-product of user responses to emerging technologies. Identifying both types provides a broader empirical basis for understanding creepiness in HCI.

To further interpret patterns across the corpus, we clustered papers within each of the two main categories INTENTIONAL and UNINTENTIONAL CREEPINESS into thematic topics. This allowed us to analyse not only whether creepiness was a design focus but also the contexts and mechanisms through which it emerged.

3.2.1 Intentional Creepiness. A majority of the papers (70.43%) in our corpus examined creepiness as an explicit research concern. These studies aimed to understand how creepy effects are produced, perceived, and designed for in interactive systems. We identified eight thematic clusters in this group: (i) *uncanny valley*, (ii) *human-likeness*, (iii) *robots and other autonomous technologies*, (iv) *health and vulnerable populations*, (v) *privacy and data concerns*, (vi) *voice interaction and voice assistance*, (vii) *methods*, and (viii) *miscellaneous*. Below, we present three illustrative examples to demonstrate how creepiness was intentionally investigated across different themes.

The first example belongs to the *uncanny valley* category. Kaate et al. [55] employed deepfake technology in the construction of user personas to study how near-human visual representations influence user experience. Their findings revealed that subtle imperfections in facial realism produced discomfort, shaping perceptions of empathy, credibility, and control. This directly engages with the uncanny valley hypothesis, which posits that entities that closely, but not fully, resemble humans often elicit a sense of eeriness or unease. The second example, situated in the *human-likeness* category, also investigates creepiness directly, but with a different emphasis. While uncanny valley studies often focus on perceptual thresholds of discomfort, research on human-likeness explores how degrees of resemblance influence interaction more broadly—including, but not limited to, creepiness. Shih et al. [117] examined how avatar similarity in virtual reality shaped user perceptions of eeriness and persuasiveness. Their work demonstrated that increased visual similarity could heighten feelings of unease, even as it enhanced social presence. Here, creepiness was an intentional construct under investigation, positioned within a wider analysis of how human-like qualities affect digital communication.

A third example comes from the domain of *robots and other autonomous technologies*. Yam et al. [139] investigated how intentionally dehumanising a robot—by limiting its perceived emotional capacity—could reduce feelings of eeriness. By suppressing expressive cues, the robot became more acceptable to users, suggesting that the design of emotional behaviour is a powerful lever for modulating creepy responses. This study highlights how creepiness can be purposefully addressed through fine-tuned manipulation of social and affective signalling in robotic systems.

3.2.2 Unintentional Creepiness. Our analysis revealed that some papers reported creepiness not as a central research aim, but as an unintended outcome of design or interaction (29.5%). We identified four topical emphases within this group: (i) *AI and machine learning*, (ii) *agents and bots*, (iii) *security and privacy*, and (iv) *touch and emotional interaction*. Below, we present illustrative examples from these categories. In *AI and machine learning* , creepiness often stemmed from system unpredictability. For instance, Yurman and Reddy [144] studied creative collaboration with AI-generated watercolour art. Although the focus was on creativity, the AI's opaque behaviour introduced discomfort and disrupted collaboration—highlighting how a lack of transparency can unintentionally provoke unease. An example from *touch and emotional interaction* is provided by Lee et al. [66], who developed an artificial plant to support health goals. The plant's drooping leaves signalled inactivity, but users interpreted this behaviour emotionally, describing it as sad, dying, or uncanny. Despite lacking anthropomorphic features, the artefact evoked strong affective responses, revealing how lifelike behaviours can unintentionally produce creepy effects.

3.3 Prototypes

In this section, we examine how prototypes have been employed in the study of creepy technologies, drawing on insights from our literature review. For the purpose of analysis, we treated each prototype as an individual unit of inquiry; if a paper presented multiple prototypes, we coded each one separately. This allowed us to capture the full diversity of approaches represented across the corpus. We identified two overarching categories: (i) prototypes specifically developed for the purposes of the study (`prototype development`), and (ii) prototypes that incorporated existing technologies or components as the basis for investigation (`existing technologies as prototypes`). This distinction highlights the methodological breadth of the field, spanning original design work as well as adaptive reuse of established technologies. Analysing these prototypes is essential, as they provide different perspectives on how creepiness is conceptualised, operationalised, and experienced in technological contexts. They also demonstrate how design choices influence user perception and offer concrete insights for shaping future design practices. Our analysis showcases a range of prototype forms that collectively reflect the methodological diversity of this research domain.

3.3.1 Development of Prototypes. Across the corpus, slightly more than half of the studies (58 out of 115, 50.44%) developed prototypes, ranging from low- to high-fidelity. Four papers (3.48%) combined prototype creation with the use of existing artefacts, and 16 papers (13.91%) did not utilise prototypes in their studies. For example, Yip et al. [142] conducted four workshops with children to explore the concept of 'creepy technology'. During these workshops, the children created exploratory prototypes such as sketches and scenarios based on technologies they considered creepy. To complement this, the researchers also presented YouTube videos of ten different technologies, including Pepper[5], a semi-humanoid robot, and the Amazon Alexa Kids Edition[6].

[5] https://us.softbankrobotics.com/pepper
[6] https://www.amazon.com/alexa-for-kids/bie=UTF8&node=21474972011

Regarding `prototype development`, we identified 19 prototype categories, which we grouped into five higher-level categories. The categories are listed below and are based on similarities in bodily interaction, sensory engagement, and utility (see Figure 4). In addition to type, each prototype was also coded for its `tangibility`, `sensory modality`, and `proximity to the body`.

- Low-fidelity: 'scenarios', 'personas', 'sketches', 'storyboard', 'vignette'
- Multimedia and animation: 'face expression videos', 'voice interaction', 'animation', 'digital media'
- Virtual environment and digital spaces: 'chatbots', 'VR environments', 'virtual agents', 'VR platform'
- Wearable: 'stimulating corset', 'on-body data alert', 'haptic array'
- Physical artefact: 'notification plant', 'design probe', 'robots'

3.3.2 Utilisation of Existing Technology as Prototypes. Almost one-third of the papers (37 out of 115, 32.17%) utilised existing technologies as prototypes. The most common approach involved images of existing technologies ($n = 14$). For example, Higgins et al. [45] used images of virtual humans to examine empathetic responses to high-fidelity, emotionally expressive characters. Similarly, Strait et al. [124] presented images of both humans and robots to investigate whether uncanny valley effects are better explained by perceptual mechanisms (e.g. threat detection or avoidance) or by social mechanisms (e.g. fears shaped through negative media portrayals). The second most frequent category involved digital robots ($n = 11$), including ChatGPT [54, 93], and different avatars [22, 113, 134]. In addition, some papers drew on more than one existing technology, leading to coding across multiple categories.

Across this body of work, we identified 38 distinct types of prototypes based on `existing technologies`.

- Videos: 'YouTube videos', 'videos from movies', 'videos of Atlas the robot', 'videos of different robots', and 'videos of avatars'
- Images: 'virtual influencers', 'finger-Navi', 'teleoperated bottom viper', 'robots', 'CG faces', 'static images', 'CGI avatars from games', 'artist made', 'virtual avatars', 'service robot models', 'metahumans', 'people and robots'
- Text: 'Reddit', 'comments on YouTube', 'Facebook posts', 'TTS tool'
- Physical robots: 'Mechanical Turk', 'vacuum cleaners and car', 'Kaspar the robot', 'Furhat the robot', 'Pepper the robot', 'XR mirror', 'Telenoir', 'Iwaa robot', 'Sophia'
- Digital robots: 'voice interface', 'CGI avatars from games', 'VR agent Carla', 'digital avatar of Sophia', 'ChatGPT', 'virtual agents', 'Virtual humans'

3.3.3 Sensory Modalities in Prototype Development. The analysis of 'creepy prototypes' (both intended and unintended) clustered by `sensory modality` shows clear differences in how senses were engaged across the corpus. No studies employed (i) taste or (ii) olfaction as sensory modalities in this context. Only 7 out of 99 papers with prototypes (7.07%) incorporated (iii) touch, focusing on tactile interaction. In contrast, a majority of papers (71 out of 99, 71.71%) relied on (iv) vision, underscoring the prominence of visual modalities in the design and evaluation of prototypes. A smaller proportion of studies (23 papers, 23.23%) engaged with (v) audition, primarily in combination with visual cues. Overall, the findings reveal a strong emphasis on visual sensory modalities, with auditory modalities often paired with visual ones, and a near absence of tactile modalities. Taste and olfaction were not represented at all, pointing to underexplored opportunities for expanding multisensory investigations of creepiness in interactive technologies. These patterns are summarised in Table 5.

Table 5: Distribution of prototypes by `sensory modality`. Note that some prototypes engaged with multiple modalities and some papers described multiple prototypes.

Senses	Number of papers
Visual	71
Auditive	23
Tactile	7
Olfactory	0
Taste	0

3.3.4 Proximity of Prototypes to the Body. To examine the proximity of prototypes to the body, we coded papers into five categories: (i) not applicable, (ii) in-body, (iii) on-body, (iv) immersive, and (v) third inanimate object. Prior research has emphasised that closeness to the body is highly relevant for the experience of creepiness, making this dimension a noteworthy aspect of our analysis. The category (i) not applicable was used for papers that did not report the use of prototypes, images, or videos, as well as for opinion pieces and literature reviews. A total of 21 papers (18.23%) fell into this group. Among studies that included prototypes, the most frequent category was (v) third inanimate object, applied to 74 papers (64.35%). This category refers to artefacts that users interact with indirectly—such as websites or questionnaires—or prototypes presented through images or videos without direct bodily engagement. For instance, Parviainen and Søndergaard [89] was coded as a third inanimate object because participants interacted with a voice-controlled artefact by whispering commands, rather than wearing or being immersed in the technology. The category (iv) immersive included studies on virtual environments, avatars, and virtual humans, representing 16 papers (13.91%) in the corpus. An illustrative example is Al-Zubeidi et al. [1], who developed a virtual environment to simulate the experience of walking alone at night as a woman. The category (iii) on-body, which captured wearable devices, was applied to 4 papers (3.47%). Finally, no papers in our review fell into category (ii) in-body.

4 Design Dynamics and Experiential Elements of Creepy Technology

This section presents the results of our analysis addressing the second research question:

- **RQ2**: Which design dynamics and experiential elements are linked to perceiving particular technologies as creepy?

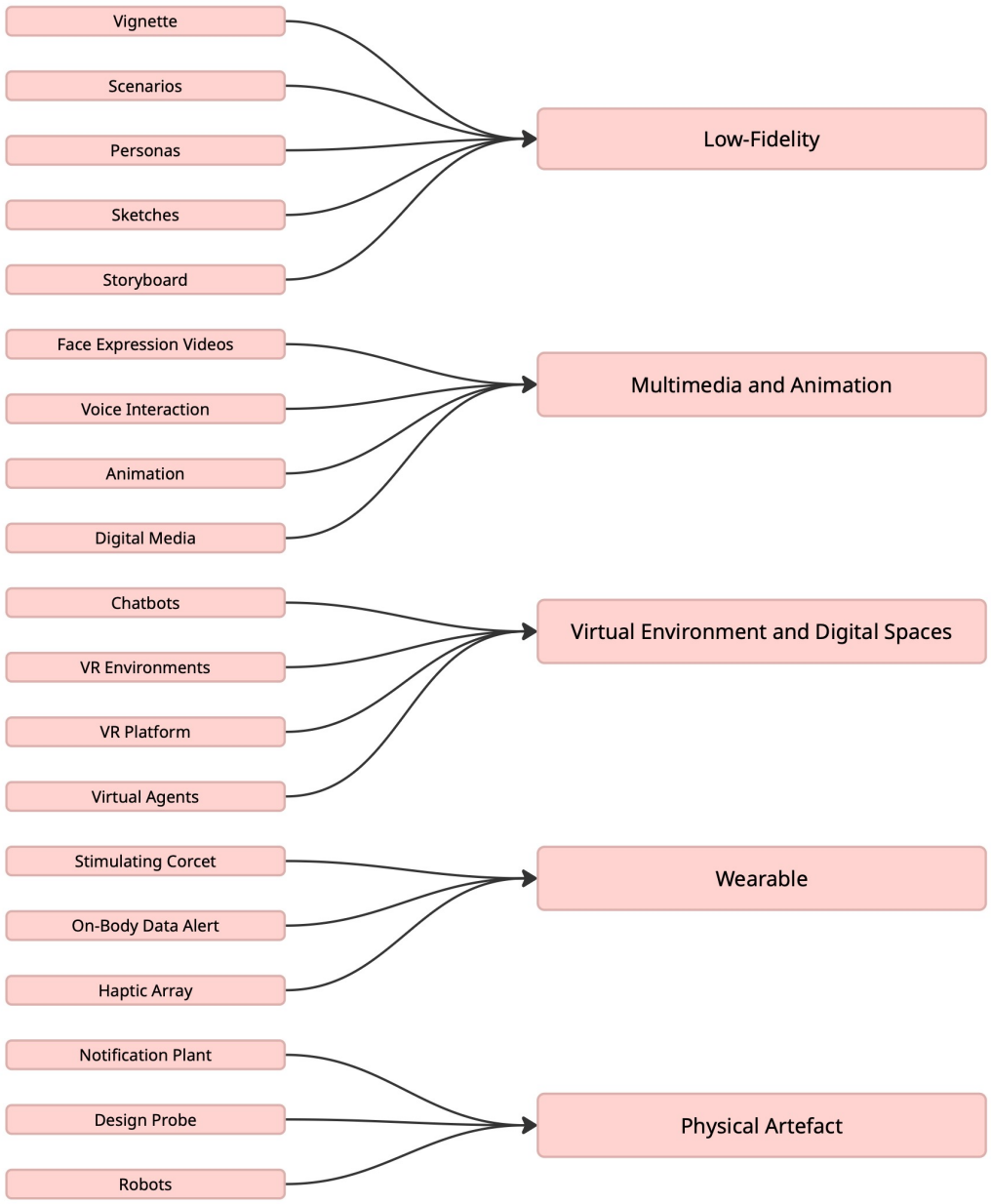


Figure 4: Categorisation of lower-level prototypes (e.g. vignettes, sketches, chatbots, haptic arrays) into higher-level groups: LOW-FIDELITY, MULTIMEDIA AND ANIMATION, VIRTUAL ENVIRONMENTS AND DIGITAL SPACES, WEARABLES, and PHYSICAL ARTEFACTS.

Our findings indicate that research on creepiness can be grouped into two overarching categories, which we refer to as the *Structure of Creepiness* and *Felt Creepiness*. The first category, *Structure of Creepiness*, captures technology-related aspects that are under the control of designers, such as design dynamics and appearance elements. The second, *Felt Creepiness*, encompasses experiential aspects shaped by users' personal histories, cultural contexts, and individual perceptions. Together, these dimensions constitute the *Creepiness Framework*, as illustrated in Figure 5.

To identify these categories, we analysed how design dynamics and experiential elements were discussed across the corpus. Our approach combined affinity diagramming [69] to cluster related concepts, concept mapping [127] to examine hierarchical relationships, and iterative discussions between the first and last author to build an in-depth understanding of nuances and interrelations. Not only did the detailed analysis of prototypes play a central role, but so too did a multilayered analysis of how creepiness was defined and conceptualised across studies. Together, these methods

enabled us to show how technological features and user experiences jointly contribute to perceptions of creepiness. This analysis also shows how intentional and unintentional foci on creepiness inform both the *Structure of Creepiness* and *Felt Creepiness*. Studies in which creepiness emerged unintentionally—e.g. AI systems that produced unpredictable behaviour or artefacts such as artificial plants that evoked feelings of sadness and eeriness—illustrate how design choices can inadvertently activate elements of the *Structure of Creepiness*. Conversely, research that explicitly centred on creepiness, such as studies of the uncanny valley, explored subjective perceptions directly and highlighted the emotional responses that constitute *Felt Creepiness*. By distinguishing between INTENTIONAL and UNINTENTIONAL CREEPINESS, these findings reinforce the dynamic relationship between design factors and user experiences, offering important insights for creating technologies that minimise discomfort while fostering positive interaction.

All papers in the corpus contributed to the development of our framework, collectively enabling an integrated understanding of the interrelated dimensions of creepiness. To articulate each component of the framework, we draw on illustrative examples from across the corpus; naturally, some components are represented by fewer examples due to the uneven distribution of themes within the literature. Collectively, these examples demonstrate the breadth and scope of the framework as constructed. Taken together, the notion of *Felt Creepiness* captures factors that shape creepy experiences but lie largely outside the direct control of designers. These include aspects such as users' prior experiences, cultural backgrounds, and individual interpretations, elements that designers cannot deliberately 'design into' an interaction yet must remain attentive to when anticipating how their systems may be received. At the same time, the relationship between the components we describe as grounded in design (*Structure of Creepiness*) and those associated with felt experience is not always clear-cut. For example, although designers can influence how predictable a system appears, they cannot fully determine how individual users judge or respond to that predictability. By engaging with each design-driven aspect and experiential element in the subsequent section, we aim to further unpack where design agency ends and where user interpretation begins.

4.1 Design Dynamics: The Structure of Creepiness

Building on the distinction between *Structure of Creepiness* and *Felt Creepiness*, we now turn to the mechanisms that explain how creepiness emerges from technology (design) itself. While the previous section outlined how these two dimensions jointly shape the *Creepiness Framework*, the following section focuses on the components that can be directly influenced through design. In other words, the *Structure of Creepiness* encompasses the knowledge that interaction designers and HCI researchers have at their disposal when they want to avoid or purposefully design technology that is experienced as creepy. This knowledge is related to (i) how the CONTEXTUAL TRIGGERS, (ii) APPEARANCE ELEMENTS, (iii) INTERACTIONAL DYNAMICS, and (iv) the NARRATIVE AND INTENT, can influence the user experience.

4.1.1 ***Contextual Triggers.*** CONTEXTUAL TRIGGERS are situational factors that can activate feelings of creepiness in human-technology interactions. For example, as part of their scale development process, Woźniak et al. [138] discussed that contextual conditions, such as whether the visual appearance of a technology fits its environment, shape user perception. To illustrate, when a wearable technology has many visible wires and sensors and is used in a public space, its appearance can be perceived as inappropriate or even creepy. However, the same device may be considered acceptable and not creepy in, for instance, a medical context, where such aesthetics align with situational expectations. A prominent aspect within this category is the notion of breaking social norms, which often contributes to a sense of inappropriateness. Parviainen and Søndergaard [89] examine how whispering interactions with voice assistants evoke creepiness by violating normative expectations of voice interaction. Whispering, in this case, is perceived as socially inappropriate and unsettling, disrupting the typical flow of communication. By linking context with social expectations, this work shows how deviations from familiar norms—such as unexpected tones or behaviours—can leave users feeling socially out of place. Understanding these context-dependent responses is crucial for designers aiming to prevent discomfort and ensure that interactions remain aligned with users' expectations. Similarly, Jarsve et al. [54] hypothesise that the discrepancy between social norms and system behaviour may contribute to perceived creepiness. They suggest that an imbalance between the context of use and the responses of AI-infused chatbots can lead to this perception.

4.1.2 ***Appearance Elements.*** Specific design features play a central role in shaping users' perceptions of creepiness in technology. For instance, as shown by Kaate et al. [55], the inclusion of human-like traits can provoke a sense of eeriness, making anthropomorphism a critical consideration for designers. To illustrate, giving a robot a vaguely human appearance [31, 139] or including functions that suggest surveillance or intrusion [18, 52] can independently contribute to discomfort. The look of a device, whether described as cold, strange, or lifelike in an unsettling way, often evokes emotional reactions such as the experience of creepiness. Lee et al. [66] demonstrate how decorative robots, made to appear more human, can amplify these effects, suggesting that increased realism does not necessarily result in greater user acceptance. Equally important is how a technology behaves. Functionality that appears unpredictable [54, 93, 140], overly autonomous [18, 54, 94], or inappropriately responsive [52, 54, 93] can heighten a sense of unease. These concerns extend beyond privacy or security; even subtle behaviours can be perceived as manipulative or unsettling if they deviate from user expectations. Such reactions highlight the need for clear, predictable interactions that support a sense of control and understanding. Appearance elements describe the features of a technology that may evoke creepiness regardless of setting, whereas contextual triggers capture how the same features can be amplified or mitigated depending on the environment. thus, an unsettling design can become creepier in an incongruent context or more acceptable in a setting where it fits normative expectations. Taken together, these insights show how both the visual and behavioural aspects of a technology can contribute to its creepiness. For interaction designers, this underscores the importance

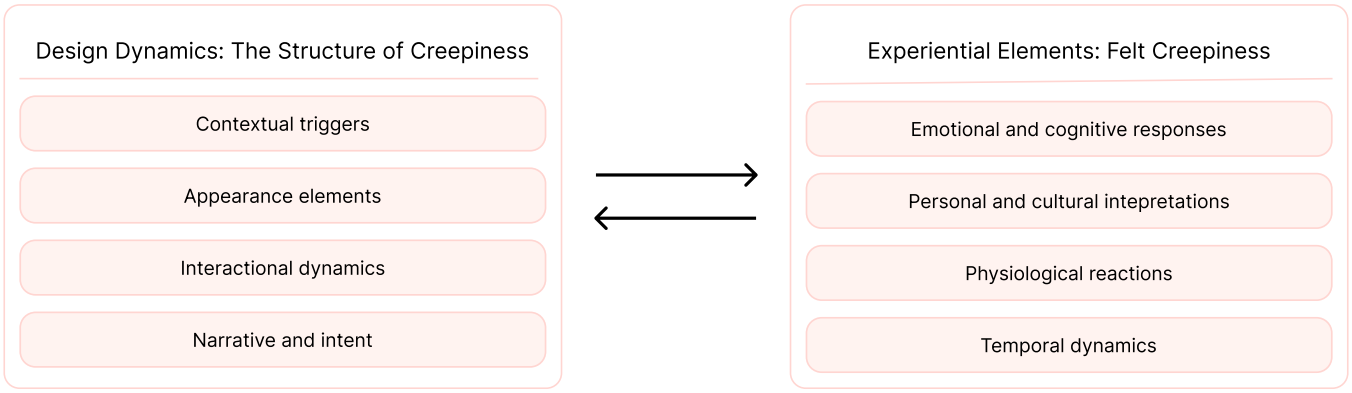


Figure 5: The *Creepiness Framework*, mapping the interplay between design dynamics (*Structure of Creepiness*) and experiential elements (*Felt Creepiness*). The *Structure of Creepiness* captures design-driven aspects such as contextual triggers, appearance elements, interactional dynamics, and narrative and intent while *Felt Creepiness* reflects the emotional and cognitive, personal and cultural, physiological, and temporal responses of users. Together, these categories demonstrate that creepiness emerges not from design features or user interpretations in isolation, but from their dynamic interplay.

of anticipating how users might interpret specific design choices, particularly when technologies exhibit human-like features or act in unexpected ways.

4.1.3 ***Interactional Dynamics.*** The way users engage with technology in real time can shape their sense of creepiness, particularly when interactions feel intrusive, ambiguous, or out of the user's control. For example, Parviainen and Søndergaard [89] illustrate how whispering responses from voice assistants can disturb users by creating a sense of unnatural intimacy, altering the expected rhythm and tone of the exchange. These dynamics become especially problematic when users cannot clearly attribute intent or boundaries to the system's behaviour. Perceived misuse plays a central role here—not only by third parties, but also in how technologies might enable unintended or manipulative outcomes. For instance, features that allow for subtle behavioural nudging or excessive data collection without clear consent may leave users feeling watched or coerced. To illustrate, Seberger et al. [109] describe how smart devices, such as speakers and cameras, can quietly collect data without users' awareness, creating an ongoing tension between interaction and surveillance. This lack of transparency fosters mistrust and contributes to a lingering sense of unease, even in otherwise mundane interactions. Beyond privacy concerns, users often respond negatively to interactions that feel artificial, overly scripted, or emotionally manipulative, especially when these behaviours are embedded in everyday devices. For example, systems that overstep expected social or functional boundaries can provoke discomfort by appearing too knowing or overfamiliar, as seen when an AI system offers unsolicited advice based on its collected data or when it fails to respect the user's stated preferences [54]. Along similar lines, insights from Yang et al. [141] indicate that the perceived depth of knowledge in virtual agents affects user trust and comfort levels. An agent that overreaches its boundaries by displaying too much knowledge or familiarity may trigger a sense of creepiness that users find unsettling. Such agents must navigate the fine line between demonstrating competence and maintaining an appropriate level of emotional distance. These examples highlight that creepiness often emerges not from a single design flaw, but from the dynamic unfolding of interaction—where timing, tone, intent, and perceived control all matter. For designers, this underscores the need to anticipate how users interpret not just what a system does, but how and when it chooses to act during lived interaction. Design choices that support user agency, establish clear boundaries, and avoid ambiguous behaviours can help reduce the risk of unsettling experiences.

4.1.4 ***Narrative and Intent.*** The INTENT of the technology and the NARRATIVE surrounding it can influence how users perceive creepiness. For instance, a narrative that suggests malicious intent can increase feelings of discomfort, while clear communication about the technology's functions can alleviate creepiness. In Yip et al. [142]'s paper, some participants discussed how the intent behind actions can make the technology seem more or less creepy. The participants pointed out that farting is generally not considered creepy because it has no evil intent; it is simply a natural bodily function that does not aim to harm anyone. In contrast, laughter can be perceived as creepy depending on the context, especially if it is used to mock or hurt someone, leading to feelings of unease among others. This highlights how both the perceived intent behind behaviours and the narratives we create around technologies can greatly influence their creepiness factor. Importantly, such cases also illustrate the interplay between the structure of creepiness (e.g. the digital sound or timing of a laugh) and felt creepiness such as emotional responses. To illustrate, a laugh might be interpreted as creepy not only because of its immediate form but also because users draw on cultural and pop-culture associations or experiences, such as the menacing laughter of villains, when making sense of technological behaviours. These broader narrative frames can therefore amplify or dampen felt creepiness, showing how structural cues and experiential interpretations co-construct the perception of creepiness.

4.2 Experiential Elements: Felt Creepiness

In contrast, *Felt Creepiness* encapsulates the subjective, emotional experiences of users as they interact with technology. This aspect acknowledges that users' responses to technology are not uniform; instead, they are deeply influenced by individual backgrounds, cultural contexts, and personal experiences. *Felt Creepiness* comprises four central elements: how users' (i) EMOTIONAL AND COGNITIVE RESPONSES, their (ii) PERSONAL AND CULTURAL INTERPRETATIONS, their (iii) PHYSIOLOGICAL REACTIONS and (iv) TEMPORAL DYNAMICS shape the subjective experience of creepiness during interactions with technology. These elements are described in more detail in the following sections.

4.2.1 ***Emotional and Cognitive Responses.*** Users often react to technology with a complex mix of emotional and cognitive responses, ranging from mild unease to strong discomfort or fear. These reactions are shaped by both individual perceptions (which this section focuses on) and the specific characteristics of the interaction. Emotional responses refer to the immediate affective reactions that technologies evoke, such as eeriness, anxiety, discomfort, or a sense of being unsettled, often triggered by violations of expectations or ambiguous system behaviour. Cognitive responses, in contrast, involve users' interpretive judgments, such as assessing whether a system is trustworthy, controllable, predictable, or behaving in a way that aligns with social norms. Together, these emotional and cognitive dimensions shape how users make sense of a potentially creepy interaction.

For instance, Herder and Zhang [43] demonstrate how unexpected data collection practices can evoke strong negative emotions, highlighting how feelings of creepiness often arise from a perceived lack of transparency or control (i.e. a subsequent cognitive response). A common emotional response described across studies is a sense of eeriness, for example when technologies appear almost human but fail to behave in convincingly human ways. Sharevski et al. [115] describe how users may find machines unnerving when they seem to possess a humanlike mind, pointing to the discomfort triggered by blurred boundaries between human and machine cognition. Similarly, Scott et al. [108] show how human-like features that fall short of realism can provoke feelings of revulsion, reinforcing the role of the uncanny valley in shaping emotional reactions. Cognitively, users may question the system's intentions, credibility, or reliability, particularly when its actions cannot be clearly interpreted or predicted. Technologies that behave in unpredictable or unfamiliar ways can undermine user confidence or lead to confusion. Yip et al. [142], for example, describe how unpredictability in interactions can heighten creepiness by disrupting users' expectations and making systems feel less controllable. The sense of being caught off-guard—whether by a voice assistant's unexpected behaviour or an unfamiliar data collection process—contributes to a feeling of vulnerability. These emotional and cognitive responses are particularly useful when iterating on prototypes, as they provide early signals of which design choices may provoke discomfort or misinterpretation. Further, we envision that this could enable designers and researchers to analyse more precisely how specific elements of a system might be linked to particular reactions, helping refine both the artefact and its intended interaction dynamics. Taken together, these studies illustrate that creepiness is not only about what a technology looks like or does, but also about how users perceive it and how it makes users feel. Emotional reactions and cognitive responses such as suspicion, discomfort, and fear are not only valid responses but critical signals that something in the interaction feels off. For designers, recognising and addressing these reactions can help create technologies that feel more trustworthy and emotionally safe.

4.2.2 ***Personal and Cultural Interpretations.*** Users' feelings of creepiness are often filtered through their personal experiences, cultural backgrounds, and societal norms. The relationship between the *Structure of Creepiness* and *Felt Creepiness* is dynamic and mutual. This means that appearance elements not only shape how users feel and think about technology, but they also influence how users will interact with that technology in the future. Personal and cultural interpretations of these interactional dynamics affect how designers think about the *Structure of Creepiness*, as well as how users perceive something as more or less creepy based on their individual backgrounds and experiences. In other words, while the *Structure of Creepiness* captures design-driven cues that can be intentionally modified, personal and cultural interpretations highlight how these cues are understood, reframed, or even rejected by different users. Hence, design features may set the stage for creepiness, but it is users' culturally shaped interpretations that determine whether those features actually feel unsettling in practice. This distinction underscores that even well-intentioned or 'neutral' design elements can evoke drastically different reactions depending on the norms, histories, and expectations users bring into the interaction. For example, in some cultures, a technology that uses facial recognition might be viewed as an innovative tool for enhancing security, while in others, the same technology could be seen as intrusive and creepy. This concept is mirrored in the findings of Meng-Schneider et al. [76], where concerns about smart speakers vary among users. Owners worried about the potential misuse of the device, while bystanders felt uncomfortable simply being recorded. Similarly, Shalawadi et al. [114] found that users of smart home cameras also have varying expectations and feelings regarding privacy based on their roles in shared spaces. Their research indicates that introducing physical privacy covers can alleviate fears of constant surveillance, suggesting that cultural interpretations of privacy and individual experiences are crucial for effective design and communication regarding such technologies.

4.2.3 ***Physiological Reactions.*** The experience of creepiness can elicit various physiological responses, such as increased heart rate or heightened alertness. This relationship between design and physiological reactions shows how design choices can inadvertently trigger discomfort, emphasising the need to account for user responses during the design process. For instance, Karpashevich et al. [57] highlight how the design of the *Soma Corset*, through its shape-changing capabilities, can elicit both comforting and uncanny sensations, blending physiological and emotional responses. This is particularly relevant in the context of creepiness, where physiological reactions are deeply interconnected with emotional responses: bodily signals such as tension or arousal often precede or accompany feelings of unease. Similarly, Sabinson and Green [101] investigate a soft robotic surface designed to regulate human emotions through biofeedback and guided breathing exercises.

While many users reported a reduction in stress levels, others experienced discomfort related to the uncanny qualities of the robot's interactions, illustrating the subjectivity inherent in physiological responses to design. HCI has a strong tradition of capturing physiological data, including measures such as galvanic skin response, heart rate variability, and eye tracking, to better understand user experiences. This is particularly relevant in the context of creepiness, as physiological reactions are deeply interconnected with emotional responses: bodily signals such as tension or arousal often precede or accompany feelings of unease. While these methods offer valuable insight into subtle affective states that users may not be able to articulate, the act of being physiologically monitored—especially without clear consent or feedback—can itself become a source of creepiness. Designers and researchers must therefore carefully consider how such sensing technologies are introduced and communicated, to avoid reinforcing the very discomfort they aim to study.

4.2.4 ***Temporal Dynamics.*** The evolution of the feeling of creepiness over time is another critical factor. Research indicates that users may become desensitised to creepiness with repeated interactions, or conversely, they may experience an intensification of feelings of discomfort [110]. This corresponds with Paetzel et al. [87], who suggest that as users interact with a robot over multiple sessions, perceptions of competence may quickly stabilise, while feelings of threat can fluctuate, highlighting the dynamic nature of user emotional responses. Similarly, Mazuz and Yamazaki [74] note that for some users, the discomfort associated with uncanny feelings can actually increase with repeated exposure, as subtle unsettling features become more apparent, challenging their initial perceptions. This dynamic aspect of *Felt Creepiness* is crucial for understanding user engagement over longer periods. In addition, the novelty effect plays a role in shaping these trajectories: initial encounters with unfamiliar or anthropomorphic technologies may amplify creepiness, which then diminishes as users become accustomed to the system. At the same time, for certain technologies the opposite pattern may occur, where repeated exposure reveals subtle unsettling features that only emerge after novelty wears off. These shifting responses underscore the importance of evaluating user experience not just at a single point in time, but across prolonged use. Designers must account for how perceptions of creepiness may develop or accumulate, ensuring that technologies remain trustworthy and comfortable beyond the first impression.

5 Discussion

This discussion reflects on our findings in relation to the two research questions guiding this review. The first research question (**RQ1**) asked how HCI research has investigated and represented the concept of 'creepy technology'. The second research question (**RQ2**) examined which design dynamics and experiential elements are linked to perceiving particular technologies as creepy. Our results identified two overarching categories—the *Structure of Creepiness* and *Felt Creepiness*, that together explain how technological features and user experiences jointly shape perceptions of creepiness. The *Structure of Creepiness* encompasses the design-driven aspects under the control of designers, while *Felt Creepiness* captures the subjective, cultural, and temporal experiences of users. This distinction provides a framework for understanding how intentional and unintentional design choices elicit discomfort and how user perceptions evolve over time.

Taken together, these findings provide a comprehensive overview of how creepiness has been conceptualised in HCI research and clarify the mechanisms through which it is experienced. In the following sections, we discuss the implications of this work for advancing HCI research and for guiding design practice.

5.1 Future Research on Creepiness in HCI

Based on our analysis we identified several gaps and opportunities for advancing the study of creepy technology in HCI. Much of the existing literature concentrates on established themes such as the uncanny valley and privacy concerns. While these remain important areas of inquiry, this focus has led to a relatively narrow conceptualisation of creepiness. Less attention has been paid to everyday technologies—such as self-driving cars [79], traffic lights [51], or headphones [34] that can also evoke discomfort. Exploring these domains **could broaden our understanding of how creepiness arises beyond contexts that are already relatively well understood**.

A general challenge in HCI, which is also reflected in the findings from our literature review but nevertheless crucial to reiterate, is that cultural background, technological familiarity, and demographic differences remain underexplored. To illustrate, while some studies have investigated children's perceptions [124, 142], most research focuses on adults in Western contexts. **Given that creepiness is a subjective phenomenon shaped by personal histories and cultural norms, examining more diverse user groups is essential.** Without such inclusivity, HCI risks developing design guidelines that fail to resonate with a broader population. Concurrently, extending the methodological repertoire can deepen and enrich insights, given that, with some notable examples [119], HCI research on creepiness has relied heavily on interviews, questionnaires, and experimental paradigms grounded in psychology. These approaches have produced valuable findings but often overlook the rich, situated, and subjective experiences that define creepiness. Expanding the methodological repertoire to include approaches such as autoethnography [56], phenomenology, and cultural probes could provide more nuanced perspectives. In addition to that, utilising mixed method approaches that enable the combination of physiological measures—such as galvanic skin response, heart rate variability, or eye tracking—with qualitative accounts of lived experience can offer a more holistic understanding of how creepiness is embodied, perceived, and interpreted across different contexts and over time. In line with that, we found that only a few studies have explored creepiness longitudinally, even though experiences often shift with repeated exposure. The novelty effect, for instance, may amplify unease initially but dissipate over time. Conversely, subtle unsettling features may emerge only after the novelty wears off. Longitudinal designs are therefore necessary to disentangle transient discomfort from enduring perceptions. Moreover, such studies can shed light on how contextual factors influence creepiness over time, offering a more comprehensive account of how interactions unfold in real-world use.

On another note, given the increasing interest in understanding the research landscape of creepy technology, it is promising

that adequate measurement instruments have recently been developed [90, 138]. However, given the simultaneous and **pressing need for more theory in HCI [85], we envision that our framework of the *Structure of Creepiness* and *Felt Creepiness* can serve as a starting point for further theoretical development—studying their interplay and how it may be reflected in specific design characteristics and experiential qualities.**

Together, these points suggest a research agenda that is broader in scope, more inclusive in populations, and more diverse in methodology. We envision that by expanding both the conceptual and empirical horizons of creepiness research, HCI can deepen its understanding of how and why technologies are perceived as creepy.

5.2 Utilising Creepiness in Interaction Design

This section discusses how the **concept of creepiness** can inform interaction design, focusing on how designers might navigate, anticipate, and even strategically employ unsettling experiences in the design of interactive systems. For design practice, the distinction between the *Structure of Creepiness* and *Felt Creepiness* provides a valuable framework. The *Structure of Creepiness* helps designers to identify and evaluate features—such as anthropomorphism, realism, interaction modalities, or data practices—that may inadvertently elicit discomfort. *Felt Creepiness* reminds us that user perceptions are shaped by individual, cultural, and situational contexts. Designing with both perspectives in mind allows for more human-centred and ethically attuned technologies.

To illustrate, several aspects of interaction design call for critical reflection, particularly where design decisions may unintentionally elicit feelings of creepiness or discomfort. Anthropomorphism and near-human realism can provoke eeriness, as seen in uncanny valley studies [3, 47, 118]. Interaction modalities, such as whispering by voice assistants, may disrupt social norms and create discomfort even when technically functional [89]. Privacy and security also remain central concerns, as opaque or intrusive data practices consistently heighten mistrust and unease [76, 82]. We therefore recommend that designers approach these aspects not only from a technical perspective but also through the lens of user experience and emotional response. Contextual appropriateness is another critical factor. Technologies perceived as benign in one setting may feel unsettling in another. For example, facial recognition may be seen as enhancing safety in some contexts but as intrusive in others. Designers need to account for these variations by involving diverse users in the design process and evaluating systems across multiple contexts.

Based on our analysis, we identified a gap in the utilisation of creepy technology as a conceptual and methodological resource in design practice. Specifically, **creepiness holds potential as a form of provocation that can be productively employed within critical and speculative design to foster reflection, surface assumptions, and challenge normative expectations.** The nuances identified in our definitions offer a shared vocabulary, or lingua franca, for design experiments, particularly given prior work showing that users' lived experiences often diverge from how they articulate those experiences when interacting with technology [53]. Unpacking these discrepancies, how creepiness is felt, expressed, and manifested in design contexts, opens up new opportunities for interactive artefacts that intentionally leverage friction as a strategy for engagement and critical inquiry.

Another area that remains underexplored in both design and empirical research is the role of different senses in shaping the experience of creepiness. While much attention has been given to visual and auditory cues, such as appearance or voice, there has been relatively limited engagement with the tactile, olfactory, and proprioceptive dimensions of interaction. This gap represents an opportunity for further exploration, especially considering the interplay between sensory modalities, contextual demands, and cultural interpretations. For instance, haptic feedback that mimics human touch [48] may be experienced as comforting in one culture and as invasive in another. Likewise, subtle changes in sound or scent may influence emotional responses in ways that are deeply context-dependent. **Expanding the sensory scope of both design and evaluation practices could yield richer insights into how creepiness is felt and how it might be better anticipated or mitigated.** To summarise, these insights point to the need for design practices that actively balance novelty, functionality, and user comfort. We hypothesise that by using the *Structure of Creepiness* and *Felt Creepiness* as both a guiding framework and an interpretive lens, designers can more systematically account for users' emotional and contextual responses. In doing so, we envision that HCI can contribute to technologies that engage users meaningfully, foster emotional resonance, and support sustained, positive interactions over time.

5.3 Limitations and Future Work

We recognise that the literature review in this paper is subject to several limitations. This review was guided by the ACM Guide to Computing Literature, which indexes a broad and representative body of computing and HCI research across a range of publishers and venues. While the ACM Digital Library served as the main access point for identifying relevant papers, our search extended beyond the ACM's own publications to include work captured through the Guide's wider indexing system. As such, the scope of this review is not confined to the ACM DL but reflects the broader landscape of computing literature. The ACM Guide to Computing Literature aggregates and indexes a wide spectrum of computing and HCI publications, including work published in journals and conferences outside of the ACM, thereby offering a robust and representative overview of the field. Nevertheless, we recognise that some relevant research, particularly from adjacent disciplines such as Psychology or STS, may not be captured within this computing-oriented framework. Future work could complement this review by engaging with broader, interdisciplinary databases that index research from these fields, such as PsycINFO or Scopus, to further enrich the understanding of creepiness in human–technology interaction.

Our aim, however, was not to provide an exhaustive cross-disciplinary account, but rather to consolidate and synthesise how creepiness has been conceptualised and studied within HCI and computing literature. In that context, the chosen corpus reflects the central venues and discourses most relevant to our research questions.

In addition, this review did not include literature on dark patterns or on related constructs such as trust, comfort, and user reassurance. While these areas are undoubtedly relevant, our focus was on synthesising how creepiness itself has been conceptualised and studied within HCI. As such, we deliberately set aside adjacent but conceptually distinct themes to maintain a clear analytical focus. That said, future work could productively build on this review by exploring how manipulative design practices contribute to creepiness, or how positive experiences, such as trust and comfort, might counterbalance it. These directions offer promising opportunities to deepen our understanding of how user perceptions are shaped in both negative and positive terms. The current review lays a necessary foundation for such work by clarifying the contours of creepiness as a construct in HCI research and identifying where further investigation is most needed.

Lastly, we acknowledge that the Creepiness Framework requires further validation. Our analysis, based on our literature review, identifies the design and experiential elements that contribute to Felt Creepiness and the Structure of Creepiness, but additional work is needed to examine how these elements manifest across different technological domains and cultural contexts, and how they may interact. Future studies should therefore aim to empirically evaluate and refine the framework. Such validation could further strengthen our understanding of how insights from the framework can be translated into more intentional design practices, particularly those aimed at preventing unwanted creepy experiences with technology, as well as informing critical, reflective, or provocative design practices that intentionally engage with creepiness.

6 Conclusion

In this paper, we conducted a systematic literature review of 115 papers published between 2019 and May 2025 to investigate how 'creepy technology' has been represented in HCI research and the design dynamics associated with it. Our findings led to the development of the *Creepiness Framework*, which distinguishes between the *Structure of Creepiness* and *Felt Creepiness*. This framework is informed by insights into design choices, the prototypes designed and utilised in the literature, and the varying definitions related to creepiness across studies. Overall, these results provide a clearer understanding of the complexities surrounding creepiness in technology and its implications for design practices.

Acknowledgments

An AI language model, the built-in spellchecker of Overleaf, and Grammarly were used to rectify spelling errors and obtain valuable suggestions aimed at enhancing the quality of our writing. All ideas and content remain the authors.

Paweł W. Woźniak is supported by an endowment from TU Wien. Barbara Sienkiewicz is funded by the National Science Centre, Poland under the OPUS call in the Weave programme — project 2021/43/I/ST6/02489. Jasmin Niess is supported by the Design for Democracy project, funded by the University of Oslo Democracy Initiative.

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A Appendix

The appendix contains supplementary information pertinent to our systematic literature review. It includes: (i) trends in publication outlets in which the research has been disseminated; (ii) an overview of the various research contributions identified within the reviewed papers; (iii) a summary of the research methods employed across the studies; and (iv) a description of participant characteristics and sample sizes utilised in the research. This comprehensive information aims to enhance the reader's understanding of the context and scope of the findings presented in the main text.

A.1 Publication Trends & Publication Outlets

The year 2024 emerged as the most productive, with a total of 22 publications concentrating on 'creepy technology'. The yearly distribution of these published papers is illustrated in Figure 6. The chart suggests a potential increase in papers focusing on creepiness, as indicated by the peak in 2024, with the screening concluding in May 2025.

Among the 115 papers, 95 papers (82.61%) were full papers, while 19 papers (16.52%) were either extended abstracts or short papers. The least common types of work were pictorials (1 paper, 0.87%). Zero works-in-progress were analysed.

A.2 Research Methods

As shown in Table 8 (`research approach`), most papers conducted QUANTITATIVE research (47.83%), followed by MIXED METHODS (27.83%) and QUALITATIVE research (13.91%). Additionally, 12 papers (10.43%) were categorised as NOT APPLICABLE, as they were OPINION pieces, only described the development process, or were LITERATURE REVIEWS.

Among the methods used (`data collection methods`), QUESTIONNAIRES (63.47%) were the most common. This was followed by INTERVIEWS (13.91%), and EXPERIMENTS (6.9%). One recurring aspect across various methods—including interviews, workshops, OBSERVATIONS, user studies, and some questionnaires—was that the specific location for conducting the studies was often unspecified. With regards to `study duration`, with one exception, all studies were CROSS-SECTIONAL. Only two LONGITUDINAL studies were conducted [10, 74]. Karpashevich et al. [57] combined both longitudinal and cross-sectional, with the study lasting one week.

1.3 Participant Characteristics & Sample Sizes

Regarding the `types of participants`, most studies (61 out of 115, 53.04%) involved predominantly ADULTS. 21 papers (18.26%) of the papers described their participants as STUDENTS or undergraduates. Five studies [10, 21, 74, 124, 142] (4.35%) included CHILDREN as participants. One paper (0.87%) mentioned AI as a participant. The remaining papers (30 papers, 26.09%) were classified as NOT APPLICABLE (13 papers) or UNSPECIFIED (17 papers).

1.4 Research Contribution

As shown in Table 7, empirical contributions dominate the corpus with 97 papers (84.3 %), followed by artefact contributions (34 papers, 29.5 %), theoretical contributions (15 papers, 13 %), and methodological contributions (10 papers, 8.7%). Opinion contributions account for 6 papers (5.22%), while survey contributions are represented by 3 papers (2.61%). The absence of dataset contributions indicates a potential area for further exploration and development.

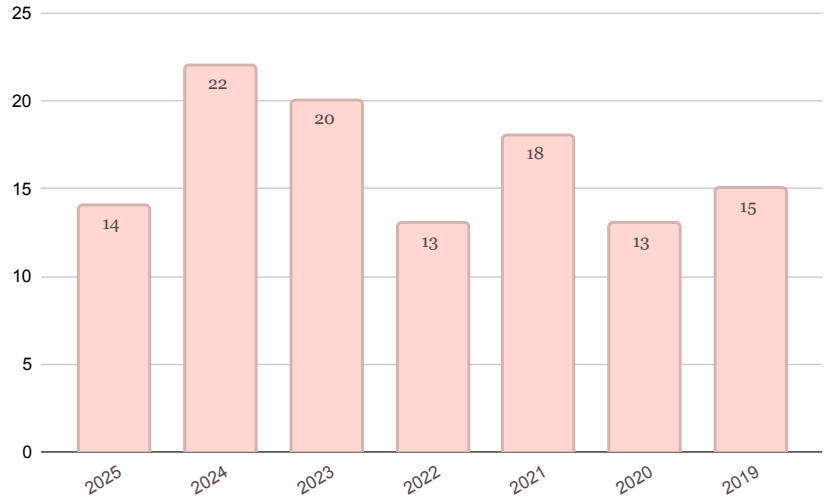


Figure 6: Publication year timeline of our corpus, displaying the distribution of included papers from 2019 to May 2025. Although the timeline shows a slight increase in the number of publications in the years 2023 and 2024 regarding the study of creepy technology in the field of HCI, there are no visibly drastic changes in the number of publications addressing 'creepy technology' over this period.

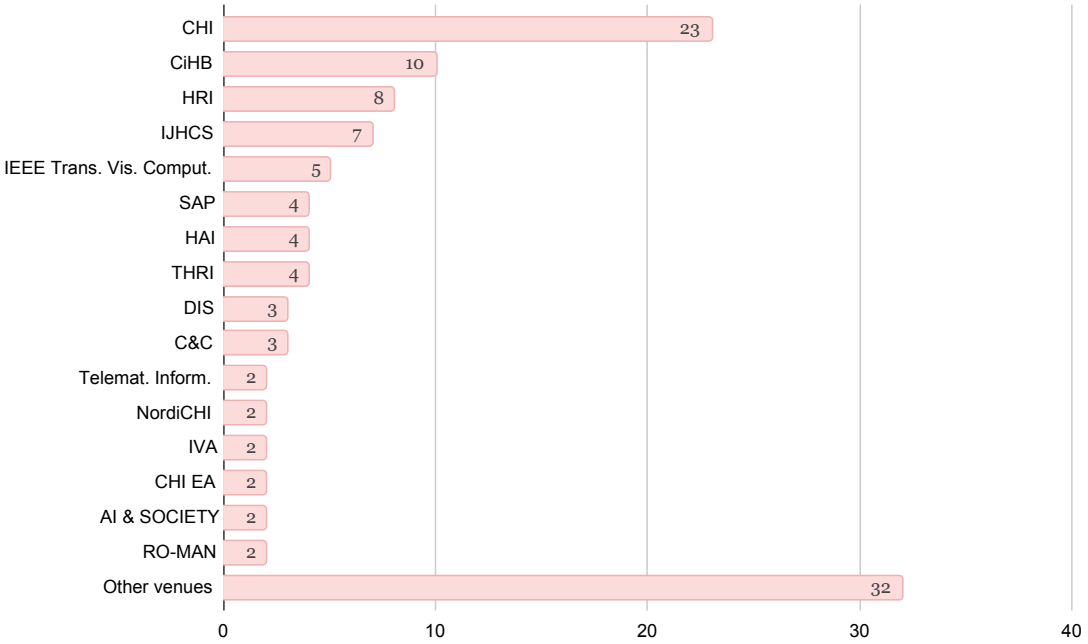


Figure 7: Distribution of the publication venues of the included papers in our corpus across HCI venues. Table 6 provides an overview of the used acronyms for each of the venues. Note that venues with only one paper are included in the 'other venues' category. These include TOCHI, MUM, and TEI, amongst others. For a comprehensive list of all venues, please refer to Table 6.

Table 6: Distribution of the publication venues of the papers included in the corpus, including the acronyms for each venue. Our corpus included 32 records which were single cases of certain publication venues.

Publication venue	# Papers
ACM CHI Conference on Human Factors in Computing Systems (**CHI**)	23
Computers in Human Behavior (**CiHB**)	10
ACM/IEEE International Conference on Human-Robot Interaction (**HRI**)	8
International Journal of Human-Computer Studies (**IJHCS**)	7
IEEE Transactions on Visualization and Computer Graphics (**IEEE Trans. Vis. Comput. Graph.**)	5
ACM Symposium on Applied Perception (**SAP**)	4
International Conference on Human-Agent Interaction (**HAI**)	4
ACM Transactions on Human-Robot Interaction (**THRI**)	4
ACM Designing Interactive Systems Conference (**DIS**)	3
ACM Creativity & Cognition (**C&C**)	3
Telematics and Informatics (**Telemat. Inform.**)	2
Nordic Conference on Human-Computer Interaction (**NordiCHI**)	2
ACM International Conference on Intelligent Virtual Agents (**IVA**)	2
Extended Abstracts of the CHI Conference (**CHI EA**)	2
AI & Society	2
IEEE International Conference on Robot & Human Interactive Communication (**RO-MAN**)	2
International Conference on Tangible, Embedded, and Embodied Interaction (**TEI**)	1
ACM Transactions on Computer-Human Interaction (**TOCHI**)	1
International Conference on Digital and Interactive Arts (**ARTECH**)	1
International Conference on Automotive User Interfaces and Interactive Vehicular Applications (**AutomotiveUI**)	1
ACM Symposium on Virtual Reality Software and Technology (**VRST**)	1
International Conference on Mobile and Ubiquitous Multimedia (**MUM**)	1
International Conference on PErvasive Technologies Related to Assistive Environments (**PETRA**)	1
SIGGRAPH Asia (**SA**)	1
International Workshop on Personalization and Recommendation on the Web and Beyond (**ABIS**)	1
IEEE/ACM International Workshop on Bots in Software Engineering (**BotSE**)	1
ACM Recommender Systems Conference (**RecSys**)	1
Augmented Human International Conference (**AH**)	1
ACM Transactions on Applied Perception (**TAP**)	1
International Conference on Multimodal Interaction (**ICMI**)	1
Mexican International Conference on Artificial Intelligence (**MICAI**)	1
New Security Paradigms Workshop (**NSPW**)	1
ACM Web Science Conference (**WebSci**)	1
International Journal of Information Management (**IJIM**)	1
International Conference on Human-Machine Interaction (**ICHMI**)	1
International Conference on Movement and Computing (**MOCO**)	1
IEEE Transactions on Haptics (**IEEE Tran. Haptics**)	1
Proceedings of the Association for Information Science and Technology (**Proc. Assoc. Inf. Sci. Technol.** or **ASIS&T**)	1
Proceedings of the ACM on Computer Graphics and Interactive Techniques (**Proc. ACM Comput. Graph. Interact. Tech.**)	1
Journal of Robotics	1
The Visual Computer	1
Universal Access in the Information Society (**UAIS**)	1
USENIX Security Symposium (**USENIX Security**)	1
Brain Informatics (**BI**)	1
Social Robotics	1
Future Generation Computer Systems (**Future Gener. Comput. Syst.** or **FGCS**)	1
Intelligent Service Robotics (**Intell. Serv. Robot.**)	1
Electronic Commerce Research and Applications (**ECRA**)	1

Table 7: Distribution of the types of research contributions in the corpus, categorised as empirical, artefact, methodological, theoretical, dataset, survey, and opinion contributions. Note that papers could be coded multiple times for different contribution types; therefore, the total number of papers in our corpus is not equal to the total number of papers listed in this table.

Contribution type	Number of papers
Empirical	97
Artefact	34
Theoretical	15
Methodological	10
Opinion	6
Survey	3
Dataset	0

Table 8: This table presents a distribution of the research approaches used in the corpus, categorised as qualitative, quantitative, mixed methods, and not applicable.

Research approach	Numbers of papers
Qualitative	16
Quantitative	55
Mixed methods	32
Not applicable	12